

MxDiffusion: A Physics-Aware Maxwell's Law-Guided Diffusion Model Strategy for Inverse Photonic Metasurface Design

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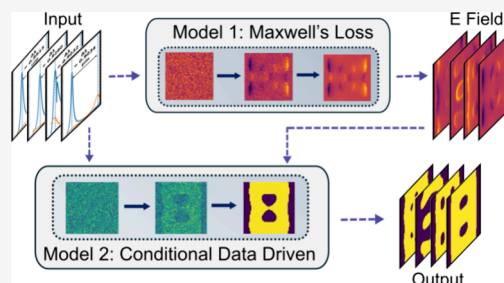
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ABSTRACT: We introduce MxDiffusion, a hybrid physics- and data-driven diffusion-based framework that enables the efficient and highly accurate generation of photonic structures from target optical properties. The improved accuracy is achieved through a two-stage generation strategy, in which the first diffusion model is explicitly trained with Maxwell's equation-based loss to embed physical insight directly into the inverse design process, while the second model maps the physically consistent intermediate representation to the final structural geometry with significantly higher fidelity than solely data-driven approaches. The performance of MxDiffusion is validated on two representative applications: gold pattern optimization for random spectral responses and a tunable bandpass filter design based on a phase change material. In both cases, the proposed framework consistently outperforms a conventional data-driven diffusion model benchmark, particularly for out-of-training distribution design targets and highly constrained resonance conditions. These results demonstrate the efficacy and superiority of MxDiffusion as a general physics-guided inverse design paradigm.

KEYWORDS: *Metamaterials, Nanophotonics, Diffusion Model, Inverse Design, Physics-Aware*



Machine learning (ML)-based photonic inverse design refers to the use of machine learning models to determine structural parameters that satisfy a prescribed optical response. Due to the time-consuming and inefficient nature of traditional trial-and-error approaches for designing photonic structures, machine learning-based techniques have emerged as powerful tools for exploring complex design objectives that are difficult or even impractical to achieve using trial-and-error methods.¹ ML-based inverse design of photonic structures has attracted significant attention over the past several years^{2–20} and have been adopted across many domains of optics and photonics, including nonlinear optics,^{21,22} quantum photonics,^{23–25} imaging systems,^{26–28} photonic crystals,²⁹ and metasurface design,³⁰ to name a few.

While deep neural networks (DNNs), such as fully connected networks (FCNs) and convolutional neural networks (CNNs), are well suited for forward prediction tasks,^{31,32} for example, predicting the transmission spectrum of a metasurface, inverse design problems typically require generative models, such as variational autoencoders (VAEs) and generative adversarial networks (GANs), to produce new structural configurations that satisfy specified optical response targets. A variational autoencoder (VAE) is a deep generative model composed of an encoder and a decoder, where the encoder compresses input data into a low-dimensional latent representation, and the decoder reconstructs the original data by minimizing reconstruction error during training. Once trained, the decoder can generate new samples from random latent vectors. In photonic inverse design, VAEs have been

employed as pattern generators in combination with forward prediction networks or further refined using evolutionary strategies.^{21,33,34} However, VAEs suffer from limited conditional generative capability, as they primarily generate samples through random latent-space sampling, which restricts precise control over the generated patterns to satisfy complex design objectives. GANs have become the most widely used generative framework for photonic inverse design.^{35–38} A GAN consists of a generator that produces synthetic samples and a discriminator that distinguishes between real and generated data. Through adversarial training, the generator learns to produce outputs that increasingly resemble the true data distribution. Despite their success, GANs rely solely on adversarial feedback without an explicit likelihood formulation or physical constraint, which sometimes leads to unstable training dynamics, mode collapse, and limited diversity in generated designs.³⁹

To address these limitations, physics-based techniques such as adjoint optimization and physics-informed loss functions have been integrated with machine learning models to improve inverse design performance.^{39–43} More recently, diffusion

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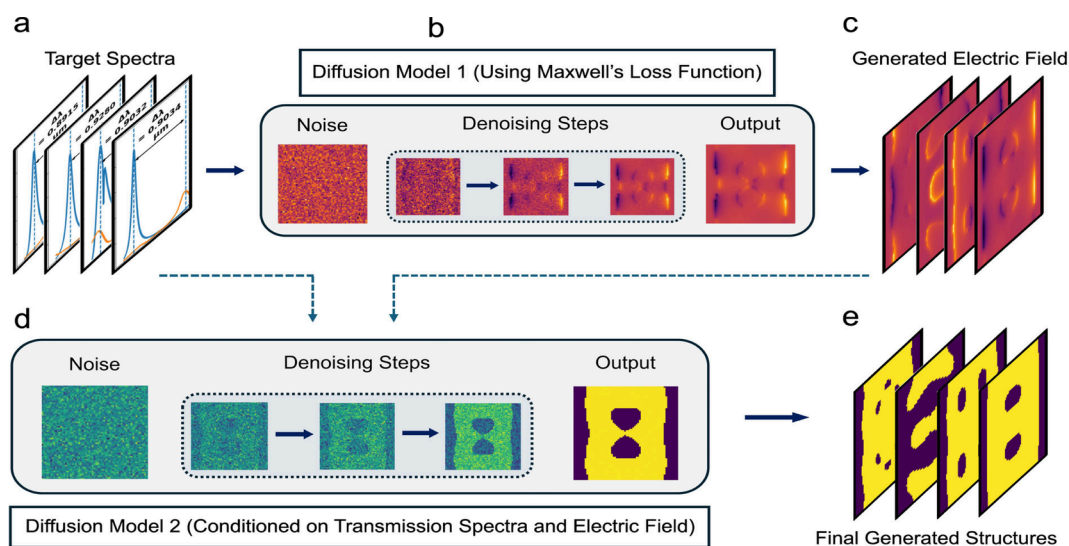


Figure 1. MxDiffusion model architecture. (a) Target transmission spectra. The objective of the model architecture is to generate 2D structure patterns that can satisfy these target spectra. (b) Diffusion model 1. This model uses Maxwell's loss to predict the electric field from the spectral data input. (c) Generated electric field pattern from the first diffusion model. (d) Diffusion model 2. This model uses both the initial target spectral data and the generated electric field as conditional inputs to produce the final structure. (e) Final generated patterns.

models have been introduced as a robust alternative for inverse design, as they decompose the generation process into a sequence of noise-prediction steps governed by a Markov chain.⁴⁴ Diffusion models have demonstrated superior performance over VAEs and GANs across a wide range of applications,^{45–52} including photonic inverse design using conditional inputs such as S-parameters and far-field spatial power distributions. Positional sinusoidal encoding is used as conditional data for training a diffusion model as well.⁵³ Recent efforts have also incorporated adjoint optimization into diffusion models to inject physical knowledge during sampling;⁵⁴ however, this approach requires real-time electromagnetic simulations to compute adjoint gradients, resulting in substantial computational overhead.

In this work, we introduce a physics-aware MxDiffusion framework, which directly integrates Maxwell's equations into the training of a diffusion model, providing physical insight intrinsically within the learning framework. This approach significantly enhances the capability of conventional data-driven diffusion models, particularly in optimizing critical boundary regions of photonic structures to meet target design specifications. Importantly, the proposed method does not require real-time simulations during sampling, and both the training and sampling costs remain comparable to those of traditional data-driven diffusion models. The efficacy of our MxDiffusion framework is illustrated through two sets of design examples. First, we consider a periodic gold nanostructure with fixed thickness and arbitrary geometrical shape patterned on glass substrates where user-defined transmission spectra serve as the design targets. In this case, the MxDiffusion framework consistently outperforms the data-driven diffusion baseline, particularly when generating structure patterns to satisfy out-of-training distribution target spectra. In the second example, we investigate the inverse design of a highly spectra-tunable bandpass filter using a phase-change material. The objective of this design task is to maximize the spectral separation between the band-pass filter peaks corresponding to the two phases of the phase-change material. The MxDiffusion framework successfully generates

structures with spectral tunability that significantly exceeds that of the best design in the training data set, whereas the solely data-driven diffusion model barely surpasses the data set's optimal performance. These results clearly demonstrate the fidelity of the proposed framework to transcend the limitations of data-only diffusion model frameworks.

A diffusion model operates through two main phases: forward diffusion and reverse diffusion,^{55–58} as shown in Figure S1. During the forward diffusion process, clean images from the data set are progressively corrupted by adding Gaussian noise over T discrete time steps until the images become nearly pure Gaussian noise. The detailed working principle of the diffusion model is explained in the Supporting Information. The forward diffusion process is entirely defined by fixed equations with predefined coefficients; so, it does not involve any learnable parameters or neural network training.

The reverse diffusion process is responsible for generating new images by progressively removing noise from a noisy image, as illustrated in Figure S1. This process starts with a highly noisy image and reconstructs a clean sample through a sequence of denoising steps. By decomposing image generation into a series of small denoising tasks, diffusion models transform a complex generation problem into multiple simpler noise-prediction problems, leading to more stable training and improved sample quality.

The key innovation of this work is the explicit integration of frequency-domain Maxwell's curl–curl wave equation (eq 2) into a diffusion-based U-Net framework for inverse nanophotonic design. In most existing machine-learning approaches for the inverse design of nanophotonic structures, the problem is formulated as a direct mapping between a design objective and a structural geometry. The input to the neural network is usually a target optical response (such as a transmission or reflection spectrum), and the output is the corresponding structural design intended to achieve that response. However, this inverse mapping is inherently nonunique: multiple distinct geometries can produce nearly identical optical spectra. As the complexity of the target response increases, learning this one-to-many-inverse mapping becomes increasingly difficult for

neural networks, often leading to unstable training, mode collapse, or physically implausible designs.

To address these challenges, this work proposes a fundamentally different inverse-design strategy, defined as the MxDiffusion framework, that incorporates Maxwell's equations directly into the training loop through a physics-based loss function. The overall architecture of the proposed physics-integrated diffusion framework for the inverse design of photonic structures is illustrated in Figure 1. Instead of attempting to map spectra directly to structural geometries in a single step, the proposed framework introduces the electric field distribution as an intermediate physical representation. In the first stage of the framework, a diffusion model is trained to predict the two-dimensional electric field distribution corresponding to a given target optical response (Figure 1a–1c). In addition to the standard Denoising Diffusion Probabilistic Model (DDPM) noise-prediction (data-driven) loss, a Maxwell's equation-based residual loss is incorporated to enforce compliance with Maxwell's equations. This physics-based constraint ensures that the predicted electric fields are physically realizable and eliminates spurious or nonphysical field patterns that might otherwise satisfy the data-driven loss alone. This step constrains the inverse problem and significantly reduces the space for the admissible solutions. Moreover, electric field patterns tend to exhibit smoother spatial variations, particularly near material boundaries, compared with binary or sharply defined structural geometries. This smoother behavior further simplifies the learning task. In the second stage, a separate diffusion model is employed to generate the final structural geometry (Figure 1d and 1e). In this step, both the target transmission spectrum and the predicted electric field are used as conditional inputs. The electric field distribution is strongly correlated with the underlying geometry, making it a comparatively straightforward task for the diffusion model to accurately predict electric fields rather than to directly infer the final structure in a single step.

For each training batch, the data-driven loss follows the standard DDPM formulation. Let the original two-dimensional electric field be denoted by E_0 . During training, a diffusion time step $t \in [0, T]$ is randomly sampled. Using eq (S3), a noisy version of the electric field E_t is generated by adding Gaussian noise $\varepsilon \sim \mathcal{N}(0, I)$. The diffusion model then predicts the noise $\hat{\varepsilon}$ conditioned on the noisy field E_t , the time step t , and the spectral conditional input. The data-driven loss (DDPM loss) is defined as the mean squared error between the true noise and the predicted noise:

$$L_{\text{DDPM}} = \|\varepsilon - \hat{\varepsilon}\|^2 \quad (1)$$

This loss encourages accurate noise prediction across all diffusion time steps. In addition to the DDPM loss, a physics-based constraint is imposed through Maxwell's loss. The frequency-domain Maxwell's curl–curl wave equation is defined as

$$\nabla \times (\mu_0^{-1} \nabla \times E) - \omega^2 \varepsilon_r E = 0 \quad (2)$$

where μ_0 is the magnetic permeability of free space, ε_r denotes the material permittivity, and E is the electric field. From eq 2, Maxwell's loss can be defined as

$$L_{\text{Maxwell}}^{(E)} = \frac{1}{N} \sum_{n=1}^N \left\| \nabla \times (\mu_0^{-1} \nabla \times \hat{E}^{(n)}) - \omega^2 \varepsilon_r \hat{E}^{(n)} \right\| \quad (3)$$

where $\hat{E}^{(n)}$ is the predicted electric field for the n th sample in the batch, ω is the angular frequency, and N is the batch size. Physically realizable electric fields satisfy eq 2 exactly; therefore, Maxwell's loss ideally evaluates to zero for realistic electromagnetic fields.

However, the diffusion U-Net architecture predicts noise rather than the electric field directly. Consequently, Maxwell's loss cannot be applied directly to the network output. To enable the incorporation of Maxwell's loss, eq (S3) is rearranged to express the original electric field in terms of the noisy field and the noise:

$$E_0 = \frac{E_t - \sqrt{1 - \bar{\alpha}_t} \varepsilon}{\sqrt{\bar{\alpha}_t}} \quad (4)$$

This relation implies that if the noisy electric field E_t and the noise are known, then the original field E_0 can be reconstructed. During training, the original electric field E_0 and time step t are first used to generate the noisy field E_t via eq (S3). The noisy field E_t , together with the spectral conditional input and time step t , is then passed through the diffusion model to predict the noise $\hat{\varepsilon}$. Substituting the predicted noise $\hat{\varepsilon}$ into eq 4 yields a reconstructed electric field $\hat{E}_0$. Since $\hat{\varepsilon}$ approximates the true noise, $\hat{E}_0$ deviates from the ground-truth field. The Maxwell loss (eq 3) is then evaluated on $\hat{E}_0$ and backpropagated through the network, enforcing physical consistency during training.

For large time step values, the predicted noise becomes increasingly uncertain, leading to large reconstruction errors and correspondingly high Maxwell's loss. Therefore, Maxwell's loss is applied only for small time-step values, where the reconstructed electric fields remain physically meaningful. In practice, the model is trained in parallel using uniformly sampled time steps $t \in [0, T]$ for DDPM loss and restricted time steps $t < 100$ for Maxwell's loss. The range of time steps used for incorporating Maxwell's loss was determined empirically through trial-and-error, and the range yielding the best performance was selected. This combined training strategy effectively balances data fidelity with physical realism. The complete training procedure is summarized in Table 1.

The first case study presented in this work focuses on the inverse design of gold nanostructures patterned on a silica substrate to achieve arbitrary on-demand transmission spectra. The thickness of the gold layer is fixed at 40 nm, and the unit-cell periodicity is set to 320 nm in both the lateral dimensions, as illustrated in Figure 2a. The training data set consists of 15,000 randomly generated gold patterns. A normally incident electromagnetic wave spanning wavelengths from 0.5 to 1.8 μm is used as the excitation source. In addition to the transmission response, the proper collection of electric field distribution plays a central role in the proposed framework. Specifically, the electric field E is collected at a plane positioned 5 nm below the bottom of the gold pattern, inside the glass. Because the electric field is sampled in close proximity to the metallic structure, its spatial distribution closely resembles the underlying geometry (as shown in the Figure 2b inset). At the same time, the electric field exhibits smoother boundaries and inherently satisfies Maxwell's equations, making it a physically meaningful and well-behaved intermediate representation.

A mesh size of 10 nm is used for electric field sampling, resulting in a 33×33 electric field profile. Consequently, each data sample in the data set comprises three components: a 33

Table 1. Complete Training Procedure (Algorithm) of the Physics-Guided Diffusion Model

	Step	Description	Explanation
Normal DDPM Training Stage	Repeat		
	1	$x_0 \sim q(x_0)$	Sample clean data
	2	$t \sim \text{Uniform}(1, \dots, T)$	Sample time step
	3	$\epsilon \sim \mathcal{N}(0, I)$	Sample Gaussian noise
	4	$x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$	Compute noisy input
	5	$L = \ e - \epsilon_\theta(x_t, t)\ ^2$	Compute DDPM loss
Physics-Guided Stage	6	$\theta \leftarrow \theta - \eta \nabla_\theta L$	Update model parameters
	7	$(x_0, c) \sim q(x_0, c)$	Layout mask + target spectrum
	8	$t \sim 75\% \text{ from } [0, 20], 25\% \text{ from } [21, 100]$	Biased time-step sampling
	9	$\epsilon \sim \mathcal{N}(0, I)$	Gaussian noise
	10	$x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$	Forward diffusion
	11	$\hat{\epsilon} = \epsilon_\theta(x_t, t, \text{embed}(c))$	U-Net noise prediction
	12	$\hat{x}_0 = \frac{x_t - \sqrt{1 - \alpha_t}\hat{\epsilon}}{\sqrt{\alpha_t}}$	DenoiSED layout
	13	$L_{\text{Maxwell}}(E) = \ \nabla \times (\mu_0^{-1} \nabla \times E) - \omega^2 \epsilon_r E\ $, where $E = \hat{x}_0$	Maxwell's loss
	14	$\theta \leftarrow \theta - \eta \nabla_\theta L_{\text{Maxwell}}(E)$	Gradient descent on Maxwell's loss
	Until converged		

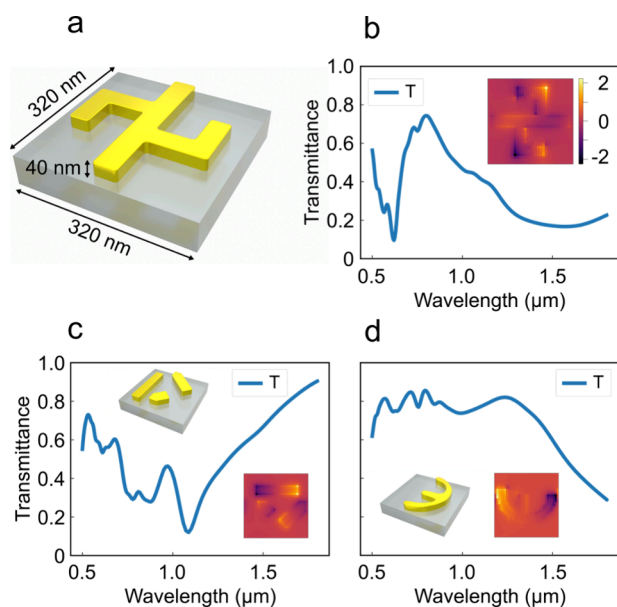


Figure 2. Data set generation process for design problem 1. (a) Unit cell consisting of the gold pattern on a silica substrate. (b) Corresponding transmission spectra for the structure in part (a). The inset represents the electric field collected just below the gold–silica interface inside the substrate. (c, d) Simulated spectra of two representative gold patterns from the training data set. The gold patterns are shown in the insets, highlighting the complexity and diversity of the structural designs. The associated two-dimensional electric-field distributions are also shown as insets.

$\times 33$ electric field distribution, a 120-point transmission spectrum, and a 64×64 binary structure pattern. Two representative examples of the training data set are shown in

Figure 2c and 2d. The structures are intentionally designed to be complex and highly arbitrary in order to demonstrate the expressive capability and robustness of the proposed Maxwell-guided diffusion framework. The corresponding transmission spectra for these structures are also shown in Figure 2c and 2d, with the insets displaying the associated two-dimensional electric field distributions at the collection plane.

The electric field is recorded at the wavelength corresponding to the minimum transmittance value in order to maximize the electric field intensity. For each structure–field pair, it can be visually observed that the electric field pattern closely mirrors the original structure geometry (Figure 2a–2d). This strong correlation implies that once the electric field distribution is known, reconstructing the underlying structure becomes a significantly straightforward task. Leveraging this insight, the proposed MxDiffusion model first takes the 120-point transmission spectrum as input and generates the corresponding 33×33 electric field distribution as output. Compared with directly predicting complex structural geometries, predicting the electric field is a simpler and more stable learning task due to its smoother spatial variation and absence of sharp material boundaries. The uniformity of the electric field allows the diffusion model to focus on capturing the correct global pattern rather than being hindered by abrupt boundary transitions of poorly optimized edge features. Importantly, Maxwell's loss enforces physical consistency in the boundary regions, which is critical for accurately reproducing the target optical response.

The proposed model is a conditional diffusion U-Net with the latest developed architectures. The core features of this model are four-level U-Net hierarchy, multihead self-attention layer, coordinate channels, Classifier-Free Guidance (CFG), and Feature-wise Linear Modulation (FiLM).^{44,59–62} The model takes both time-step data and conditional spectral data and generates electric field patterns. Time-step information is encoded using sinusoidal time embeddings with an embedding dimension of 128. The spectral data are encoded using a pure convolutional encoder followed by global average pooling and a multilayer perceptron (MLP) projection. The time and spectral embeddings are concatenated and projected into a unified conditioning vector, which is injected into every residual block using Feature-wise Linear Modulation (FiLM). This mechanism allows the network to modulate feature activations dynamically based on both the diffusion time step and the target spectrum, significantly improving conditional expressiveness. The detailed model architecture is available in the Supporting Information.

Figure 3 demonstrates the effectiveness of the proposed MxDiffusion framework through a direct comparison with a conventional data-driven diffusion model that does not incorporate Maxwell's loss. To ensure a fair evaluation, both models use identical network architectures and have the same complexity. Transmission spectra are selected from the held-out validation set for the comparison task. These spectra are first used as targets for the MxDiffusion framework to generate the corresponding 33×33 electric-field distributions. The predicted electric fields, together with the target spectra, are then used as conditional inputs to the second diffusion model to generate the final 64×64 structural designs. The generated structures are subsequently simulated by using Lumerical FDTD to evaluate whether their transmission responses match the target spectra. Three randomly selected validation examples are shown for demonstration. In Figure 3a–3c, the

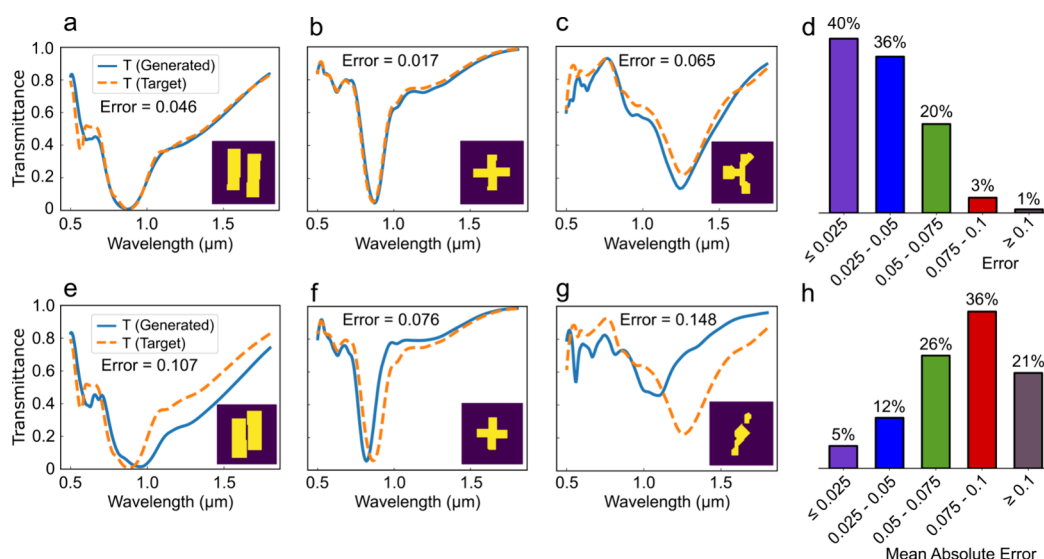


Figure 3. Performance comparison between the MxDiffusion model and solely data-driven diffusion model on the validation set. (a–c) Prediction performance of the MxDiffusion framework. Several target transmission spectra are randomly selected from the validation set and used as conditional inputs to the MxDiffusion architecture to produce the final structural designs (shown as insets). The generated structures are subsequently simulated to obtain their transmission spectra, which closely match the target spectra for most samples. (d) Error diagram for the MxDiffusion model on a validation set of 1464 samples. Most samples have very low errors. (e–g) Performance of the solely data-driven diffusion model, where the same target spectra from (a–c) are used as inputs. (h) Error diagram for the solely data-driven diffusion model. In comparison, MxDiffusion consistently achieves significantly better agreement with the target spectra than the data-driven approach.

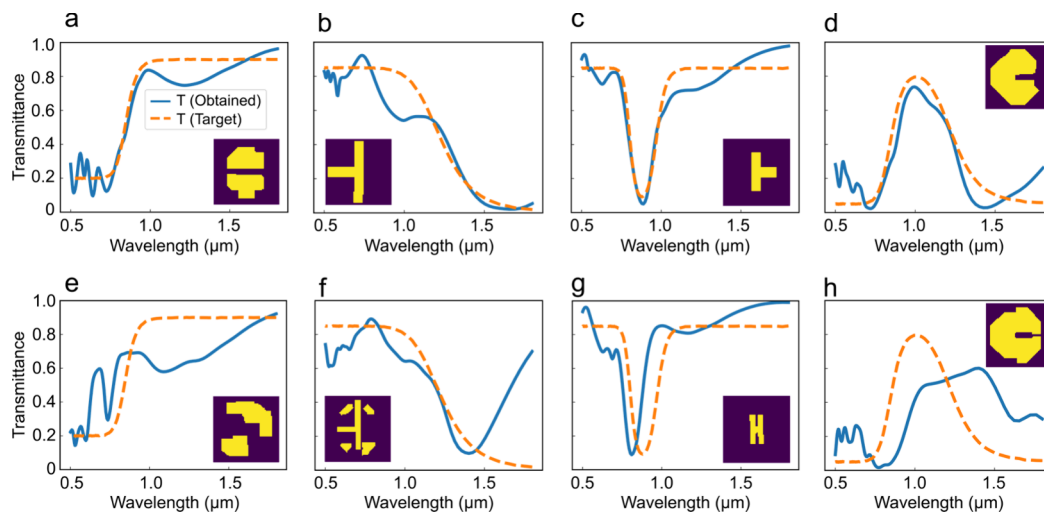


Figure 4. Performance comparison between the MxDiffusion model and solely data-driven diffusion model on out-of-training-distribution target spectra. (a–d) Target transmission spectra and simulated transmission spectra of structures generated by the MxDiffusion framework for (a) long-pass, (b) short-pass, (c) notch, and (d) bandpass filter responses. The generated patterns are shown as insets. The transmission spectra of the generated patterns accurately match the cutoff or resonance frequencies in all cases, while minor deviations in passband and stopband levels arise due to inherent physical limitations. (e–h) Corresponding results obtained using the solely data-driven diffusion model for the same target spectra, exhibiting significantly larger discrepancies from the desired responses.

target spectra and the simulated spectra of the generated structures obtained using the MxDiffusion framework are plotted together, along with the Mean Absolute Error (MAE) between the target and obtained spectra. For comparison, Figure 3e–3g present the corresponding results obtained using the data-driven diffusion model for the same target spectra.

The results indicate that the MxDiffusion framework consistently outperforms the data-only diffusion model, particularly in the target resonance regions. While both models tend to generate structurally similar patterns for a given target spectrum, the MxDiffusion framework exhibits superior boundary refinement capability, which is critical for accurately

satisfying resonance conditions. As shown in Figures 3a–3c and 3e–3g, the spectral performance of the structures generated by MxDiffusion closely matches the target spectra, especially near the resonance wavelengths. In contrast, although the data-driven model produces geometrically similar structures, its spectral responses deviate more significantly from those of the targets. Figures 3d and 3h present the performance of MxDiffusion and the data-driven model, respectively, evaluated across the entire validation set. The majority of samples generated by MxDiffusion exhibit a low mean absolute error (MAE), whereas the structures produced by the solely data-driven model consistently show significantly

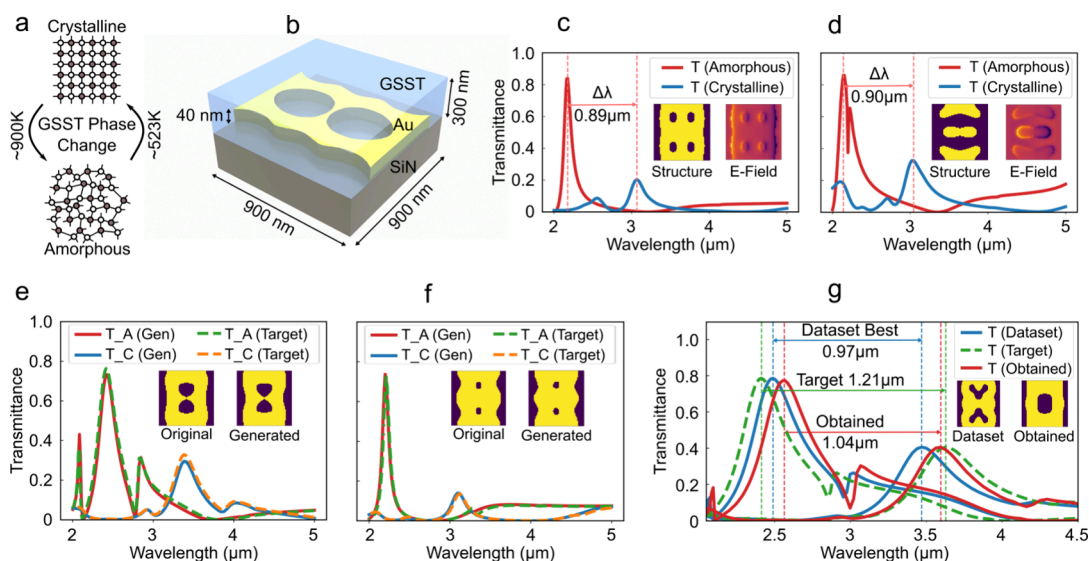


Figure 5. Schematic of the GSST-based structure and the performance of the MxDiffusion model on this data set. (a) Illustration of the phase transition of the GSST material from the amorphous to the crystalline state. (b) Unit cell structure of the GSST-based device. (c, d) Representative sample structures (inset) from the data set, where the transmission spectra for both amorphous and crystalline phases are plotted along with the resulting tunability. The corresponding electric-field distributions are also shown as insets. (e, f) Performance of the trained MxDiffusion framework on two randomly selected validation samples, showing close agreement between the target and generated transmission spectra; the original and generated structural patterns are displayed as insets. (g) Design performance of MxDiffusion for an out-of-training-distribution, highly ambitious tunability target. Here the same color is used for the two transmission curves of the two phases for a particular structure. We start from the best structure in the data set with a spectral tunability of 0.97 μm ; then the amorphous and crystalline spectra are shifted left and right, respectively, defining an aggressive target. This target spectrum is used in the MxDiffusion framework, resulting in a generated structure with a spectral tunability of 1.04 μm , an improvement of 70 nm over the best data set sample.

higher errors. These validation target spectra follow a distribution similar to that of the training data, under which MxDiffusion outperforms the solely data-driven model.

The performance of the two models is next evaluated on completely unseen target spectra that lie outside the training distribution. Figure 4 presents the results for several filter responses along with the corresponding generated structural patterns shown as insets. Figures 4a and 4e correspond to ideal long-pass filter target. After sampling both models, the MxDiffusion framework accurately captures the cutoff frequency of the target spectrum, with only minor deviations in the transmission magnitude within the passband and stopband regions. In contrast, the data-driven diffusion model exhibits substantially poorer agreement with the target response. For the short-pass filter design, as shown in Figures 4b and 4f, MxDiffusion again demonstrates superior performance. Although both models show comparable behavior in the passband, MxDiffusion achieves significantly improved suppression in the stopband region. In the notch filter design shown in Figures 4c and 4g, MxDiffusion accurately reproduces the target resonance wavelength, whereas the data-driven model fails to align the resonance correctly. Similarly, for the band-pass filter in Figures 4d and 4h, MxDiffusion consistently outperforms the baseline model in matching the desired spectral response. These results highlight the strong generalization capability of MxDiffusion beyond the training distribution. In the following design problem, a more centrally distributed data set is employed to further demonstrate the ability of the MxDiffusion framework to generate entirely new and previously unseen structural patterns.

The second design problem focuses on the realization of a highly tunable bandpass filter in the infrared spectral range

using the phase-change material Ge–Sb–Se–Te (GSST).^{63,64} As illustrated in Figure 5a, GSST can be reversibly switched between amorphous and crystalline phases, leading to pronounced changes in its optical properties and, consequently, the device transmission response. The device geometry is shown in Figure 5b. A 40 nm thick gold nanostructure is patterned on top of a Silicon Nitride (SiN) substrate, the entire structure is subsequently capped with a 300 nm GSST layer, and the unit-cell periodicity is set to 900 nm in both the lateral dimensions. An electromagnetic wave is incident from the top of the structure, and the transmission spectrum is recorded at the bottom. In this design problem, the 66×66 electric field is sampled at a plane located 1 nm below the interface between the gold pattern and the SiN substrate, inside the substrate. As in the previous case studies, each data sample consists of three components: the structural pattern image, the corresponding electric-field distribution, and the transmission spectrum.

In this study, the data set is restricted to structures that exhibit bandpass filtering behavior. The data set contains a total of 24,000 unique x–y symmetric samples. Both the amorphous and crystalline phases of the GSST are simulated for each structure, and the transmission spectra for both phases are recorded. A key characteristic of the GSST-based design is that the resonance wavelength undergoes a red shift when the material undergoes a transition from the amorphous to the crystalline phase. The difference between the resonance wavelengths in the two phases is defined as the tunability of the filter. Figures 5c and 5d present two representative examples from the data set. For each pattern, the transmission spectra corresponding to both GSST phases are plotted, and the associated gold pattern and electric-field distribution are shown in the insets. The resulting tunability for each design is

also indicated, highlighting the strong phase-dependent spectral control enabled by the GSST-based platform.

The MxDiffusion framework is trained on the GSST-based bandpass filter data set described above. In this case, each structure is associated with two transmission spectra, corresponding to the amorphous and crystalline phases of GSST. Since each spectrum consists of 120 sampling points, the total conditional input to the model is a 240-point target spectrum. As in the previous design problem, the first diffusion model is trained to predict the electric-field distribution from the target spectrum using a combination of the standard DDPM loss and the physics-based Maxwell's loss. The second diffusion model then generates the final structural pattern conditioned on the predicted electric field and target spectra. Because the electric field closely resembles the underlying structure, the second-stage prediction is relatively straightforward with the primary modeling burden handled by the first physics-driven stage.

The network architecture used for this problem is identical to that employed in the first case study, incorporating a four-level U-Net hierarchy, FiLM conditioning, a multihead self-attention block, coordinate channels, and classifier-free guidance. The total number of trainable parameters is approximately 60 million. After training, the model is evaluated on a held-out validation set. Figures 5e and 5f present two representative validation examples where the ground-truth structures and the generated structures are shown as insets. For each case, the MxDiffusion model generates the electric field from the target spectra, which is then used by the second diffusion model to produce the final structure. The generated structures are subsequently simulated using Lumerical FDTD, and the resulting transmission spectra are compared with the targets. The close agreement between the simulated and target spectra demonstrates the effectiveness of the proposed framework.

The primary objective of this design problem is to generate structures with a highly tunable spectral behavior. To establish a baseline, the structure in the data set with the highest tunability is first identified. The best tunable structure is shown in Figure 5g. This structure exhibits a tunability of 0.97 μm . A more ambitious target is then defined by manually shifting the amorphous-phase spectrum toward shorter wavelengths and the crystalline-phase spectrum toward longer wavelengths, resulting in a target tunability of 1.21 μm . Using this target, the MxDiffusion model generates a new structure, shown in the inset of Figure 5g, which achieves a tunability of 1.04 μm . Although this value does not fully reach the intentionally aggressive target, due to inherent physical constraints, it represents a substantial improvement over the best structure available in the data set. Notably, this generated structure lies outside the original data distribution, yet the MxDiffusion model successfully predicts a design with a superior performance. We also trained the model with a data-driven diffusion model, but that model cannot go beyond the best structure from the data set. These results highlight the ability of the MxDiffusion framework to extrapolate beyond existing designs and improve inverse-design performance beyond the limits of traditional data-driven approaches.

To summarize, we have introduced a physics-guided diffusion-based inverse design framework that integrates Maxwell's equations as an intermediate physical validation constraint within the diffusion process. Incorporating Maxwell's law enables the model to explicitly enforce electro-

magnetic consistency during training, significantly improving its ability to refine structural boundary regions, which play a critical role in achieving precise resonance conditions and accurately satisfying target design specifications. The proposed framework is systematically evaluated against a conventional data-driven diffusion model by using both validation samples drawn from the training distribution and completely unseen target spectra. In all cases, the MxDiffusion model consistently outperforms the solely data-driven baseline with particularly pronounced improvements observed for out-of-training-distribution design objectives. By embedding physical insight directly into the learning process, the model is able to extrapolate beyond the training data set and generate high-performance designs that are not present in the original data distribution. While the demonstrations in this work focus on two-dimensional pattern optimization, the proposed approach is naturally extensible to three-dimensional inverse structural design problems without conceptual modification, as three-dimensional electromagnetic field distributions are also governed by Maxwell's equations. Furthermore, this physics-guided diffusion strategy is broadly applicable to a wide range of photonic design tasks, including photonic crystals and multilayer metasurfaces. The design objective is not limited to spectral responses; any physically measurable electromagnetic target can be incorporated. Thus, the proposed framework provides a general and extensible solution for physics-aware inverse photonic design for a variety of optical and nanophotonic design goals.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.6c00943>.

Diffusion model overview, detailed mathematics of the forward steps of the diffusion model, detailed mathematics of the reverse steps of the diffusion model, forward and reverse diffusion steps illustration, U-net model architecture, gradual generation of electric fields from the Gaussian noise, gradual generation of final pattern from the Gaussian noise, and time-step selection and training procedure (PDF)

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Notes

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